

Ansible Course — Day 1 | Session 2

Diligence Scholarship Program

LAB 1

Inventory Setup

Duration: ~30 min · Individual · Beginner

■ Scenario

You have just joined a DevOps team. Your manager gave you two EC2 servers and asked you to set up Ansible so the team can start managing them. Organise the servers into proper groups, add variables, and confirm Ansible can reach all of them.

■ What You Will Build

- A working project folder with `ansible.cfg`
- A YAML inventory with two host groups
- Group variable files for each group
- Proof that Ansible can reach all servers

Part 1 — Project Structure

Create the project layout below on your control node.

```
~/ansible-lab/  
├── ansible.cfg  
├── inventory/  
│   ├── hosts.yml  
│   └── group_vars/  
│       ├── webservers.yml  
│       └── dbservers.yml
```

Requirements:

1. Create the folder structure shown above.
2. Create `ansible.cfg` with:
 - `inventory path · your SSH user · your private key path · host_key_checking = False`
3. Use a relative path for inventory — not an absolute path.
 - *Correct: `./inventory/hosts.yml` Wrong: `/home/abdo/ansible-lab/inventory/hosts.yml`*

Part 2 — Inventory File

Requirements:

1. Create hosts.yml with two groups: webservers and dbservers.
2. Add at least one host to each group with the correct ansible_host IP.
3. Create a parent group called production that contains both groups as children.
4. Create group_vars/webservers.yml with:

■ `nginx_port: 80 server_role: web nginx_worker_processes: 2`

5. Create group_vars/dbservers.yml with:

■ `db_port: 5432 server_role: database max_connections: 100`

■ *Use only spaces in YAML — never tabs. Always indent with exactly 2 spaces per level.*

Part 3 — Connectivity

Requirements:

1. Run the following command and confirm all hosts return SUCCESS:

```
ansible all -m ping
```

2. Run ping on the webservers group only:

```
ansible webservers -m ping
```

3. Verify your group variables are loaded correctly:

```
ansible-inventory --host web01
```

You should see `nginx_port`, `server_role`, and the other group variables in the output.

■ *If `ansible all -m ping` fails, check: Is the IP correct in `hosts.yml`? Is the SSH key path correct in `ansible.cfg`? Are there any comments (`#`) on the same line as values in `ansible.cfg`?*

■ Bonus

1. Create `inventory/host_vars/web01.yml` and override `server_role` to `primary_web`.
2. Confirm the override by running:

```
ansible-inventory --host web01
```

web01 should show `server_role: primary_web` while web02 still shows `server_role: web`.